

Cross-Lagged Network Models

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Abstract

Network theory and accompanying methodology are becoming increasingly popular as an alternative to latent variable models for representing and, ultimately, understanding psychological constructs. The core feature of network models is that individual observed items (e.g., symptoms of depression) are allowed to directly influence each other, resulting in an interconnected system of items. The dynamics of such a system give rise to emergent states of constructs (e.g., a depressive episode). Network modeling has been applied to cross-sectional data and intensive longitudinal designs (e.g., data collected using an Experience Sampling Method). Currently lacking in the methodological toolbox of network modeling is a method with the capacity to estimate network structures at the item level on longitudinal data with relatively few measurement occasions. As such, in this paper, we present a cross-lagged panel network model to reveal item-level longitudinal effects that occur within and across constructs over time. The proposed model uses a combination of regularized regression estimation and SEM to estimate auto-regressive and cross-lagged pathways that characterize the effects of observed components of psychological constructs on each other over time. We demonstrate the application of this model to longitudinal data on students' Commitment to School and Self-Esteem.

Keywords: Network Models, Cross-Lagged Panel Design, Cross-Lagged Networks.

Cross-lagged panel models (CLPMs) are common in developmental research because they allow researchers to make headway toward distinguishing the direction of causality between two or more related constructs (e.g., neuroticism, depression). Increasingly, researchers fit CLPMs in a structural equation modeling (SEM) framework, with constructs of interest modeled as latent variables underlying a set of observed measures (Finkel, 2008; Little, Preacher, Selig, & Card, 2007). In a latent variable CLPM, causal power is represented at the level of the latent construct: Constructs are presumed to affect each other over time (e.g., the latent construct ‘neuroticism’ affects the latent construct ‘depression’). Recently, network models have been presented as an alternative to latent variable models for many psychological constructs. The key feature of network models is that relations are modeled at the level of individual items rather than at the level of latent variables (Borsboom, 2008; Borsboom & Cramer, 2013; Cramer, Waldorp, van der Maas, & Borsboom, 2010; Cramer et al., 2012). For example, instead of a person’s level on a latent “extraversion” construct causing that person both to dislike people and also to dislike parties, the network modeling approach allows liking people and liking parties to be directly related (e.g., disliking people makes a person dislike parties). Thinking about constructs in a network manner is particularly relevant—from a *conceptual* standpoint—in the case where either logic/common sense, and/or empirical evidence indicates that covariance between items is better explained by a direct relation (e.g., two symptoms of major depression: insomnia $\rightarrow$ fatigue) than by a latent variable (insomnia $\leftarrow$ LV $\rightarrow$ fatigue).

In the present paper, we introduce the *cross-lagged network model*, which extends the network modeling approach to a model for longitudinal panel data (i.e., data collected on a large sample of individuals at a *few* discrete measurement occasions), which is lacking in the current literature. In a nutshell, this model will allow individual components (items) of a construct to

affect other components of the same, and other constructs, over time. As the network modeling approach is adopted rapidly in various applied fields (e.g., clinical and personality psychology; Borsboom et al., 2021; Constantini, Sarauili & Perugini, 2020; McNally, 2020) where panel data are widely available, the cross-lagged network model presents a necessary addition to the network modeling toolbox.

Cross-lagged Panel Models

In a cross-lagged panel design, two or more constructs are measured at two or more discrete measurement occasions. By regressing the set of variables at each occasion on the set of variables at the previous occasion, one can estimate the “cross-lagged” effect of each variable on the other over a particular time lag (i.e., whatever time lag separates the two measurement occasions), while controlling for the auto-regressive effect of each variable on itself. Because of the challenge with making causal inference from cross-lagged parameters (Hamaker, Kuiper, & Grasman, 2015), we interpret cross-lagged paths in terms of prediction and *not* in causal terms (a subject we will return to in the Discussion). That is, a regression path from variable X at occasion t to a variable Y at future occasion $t + 1$ suggests that a change in X predicts a change in Y over the course of that specific time lag. Such predictive relations over time generate causal *hypotheses* (Godfrey-Smith, 2009; Epskamp et al., 2018; Williamson, 2006): If X really causes Y , then regressing Y at occasion $t + 1$ on X at occasion t , controlling for Y at occasion t , should produce a non-zero regression coefficient. Finding a non-zero regression coefficient is thus a necessary but not sufficient condition for determining causality (Barnett, Barreth, & Seth, 2009).

In a traditional CLPM, the variables at each measurement occasion are scale scores (i.e., sum scores of a set of items on some psychometric scale). In recent years, it is increasingly common for developmental researchers to use latent variables to represent the constructs of

interest in a panel model (e.g., Johnson, Galambos, Finn, Neyer, & Horne, 2017). Figure 1 shows a CLPM with two latent variables and two measurement occasions. The cross-lagged paths (cl_{BA} and cl_{AB}) represent the predictive strength of each construct on the other at a later measurement occasion, controlling for the *auto-regressive* effects of each variable on itself (ar_A and ar_B).

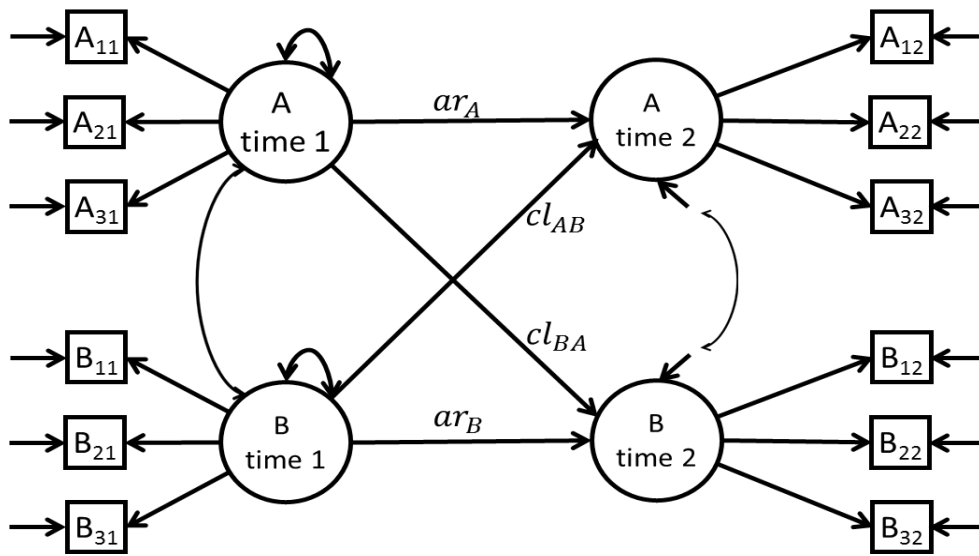


Figure 1. Cross-lagged panel model based on latent variables, with two constructs and two measurement occasions.

Hamaker and colleagues (Hamaker, Kuiper, & Grasman, 2015) introduced a cross-lagged panel model that includes a random intercept for each variable in the model. The random intercept effectively removes variance that is due to trait-like individual differences (i.e., variance that is stable within people but varies across people), so that the remaining cross-lagged and auto-regressive coefficients can be interpreted as within-person effects. It is becoming relatively common to see applications of the random-intercept CLPM (e.g., Lim, Rodebaugh, Zyphur, & Gleeson, 2016; Miklikowska, 2018).

CLPMs are most appropriate when one has data on several constructs at a few discrete measurement occasions from a large group of individuals, and when the research questions center on predictive or potentially causal effects of these constructs over time. The CLPM has known limitations that we return to later, but despite these, it continues to be a popular and useful model to investigate relations among constructs with data gathered at a limited number of measurement occasions.

Network Models

The network approach to psychological constructs (Borsboom, 2008; Cramer, Waldorp, van der Maas & Borsboom, 2010) is based on the idea that high-level attributes (disorders, traits, abilities) may arise via lower-level processes by which individual attitudes, behaviors, symptoms, beliefs, and abilities interact with each other and, as such, form dynamic systems that result in certain outcomes (e.g., an episode of major depression, being neurotic, or possessing a particular level of intelligence). Network models are an attempt to capture these lower-level processes by modeling direct relations among lower-level variables, rather than representing them as reflections of a single underlying high-level attribute (e.g., as in a latent variable model). In a typical network model, a set of items measuring a construct (e.g., the items on an extraversion scale) are represented by a set of network *nodes*. The relations between these nodes are represented by *edges*. Networks estimated from cross-sectional data (i.e., data gathered at a single measurement occasion from a large group of individuals) are typically *undirected* conditional association networks; that is, each edge represents a partial correlation between the nodes (variables) that it connects, controlling for all other nodes in the network. Conditional association networks can be estimated using l_1 -regularized regression techniques, which

estimates many edges to be exactly zero, resulting in the most “sparse” network possible given the data¹.

Figure 2 shows a regularized partial correlation network for the constructs *commitment to school* (7 items) and *self-esteem* (10 items), based on data from the Iowa Youth and Families Project (Conger et al., 2011) at the first wave of data collection (i.e., students in 7th grade; data details are presented in the “Data” section below). Green lines represent conditional positive relations among variables; thicker and darker lines represent stronger relations. The layout of nodes is chosen by an algorithm that places more strongly connected nodes closer together (Fruchterman & Reingold, 1991). Thus, one can see from the thick green lines and the placement of the variables that the items within each construct cluster together, suggesting that these constructs are conceptually coherent. Within each construct, certain pairs of items are much more strongly connected than others; for example, the items “I like school a lot” (reverse coded) and “School bores me” are more strongly related than either of those items are to the item “I try hard at school” (reverse coded). This network also reveals some connections across the two constructs, including a link between the *commitment to school* item, “I don’t do well at school” and the *self-esteem* item, “I’m a failure”. These cross-network connections generate hypotheses about the pathways by which the two constructs could influence each other over time.

Edges in a conditional association network are undirected, following from the cross-sectional nature of the data: they represent shared variance that may indicate a direct causal relation that goes in one direction or the other, a bidirectional causal pathway, the result of conditioning on a common effect, or the result of a hidden common cause that is not present in

¹ Software for estimating regularized partial correlation networks includes the EBICglasso function for Gaussian data in the R qgraph package (Epskamp, Cramer, Waldorp, Schmittmann, & Borsboom, 2012) and the R IsingFit package for binary data (van Borkulo, Borsboom, Epskamp, Blanken, Bosschloo, Schoevers, & Waldorp, 2014).

the network. An example of a hidden common cause could be that, though variables share common method variance, the type of method is not explicitly included as a variable in the model. Any potential common cause that is measured and included as a node in the network would be partialled out.

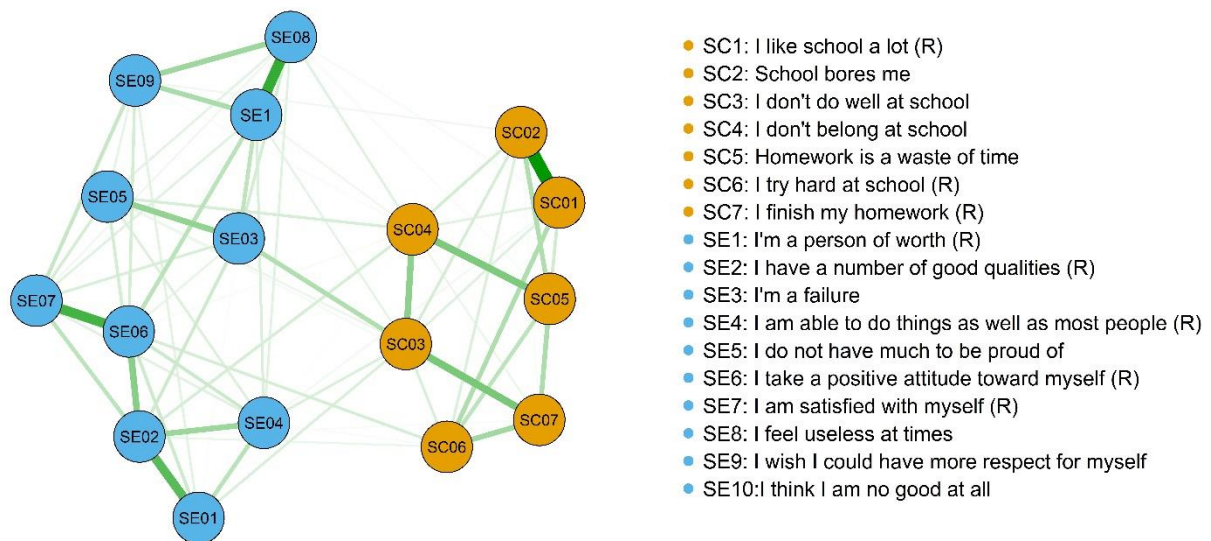


Figure 2. Regularized cross-sectional partial correlation network of Commitment to School (orange) and Self-Esteem (blue)

Many network models have also been developed for intensive longitudinal time-series data, that is, data in which a single individual or a group of individuals provide many (e.g., 100 or more) data points that are typically closely spaced in time (e.g., hours apart). Time-series network models include vector auto-regressive (VAR) models that model variables at occasion $t + 1$ as a function of those at occasion t within a single individual (e.g., Epskamp, Waldorp, Möttus, & Borsboom, 2017; Gates & Molenaar, 2012). In particular, Epskamp et al. (2017) have proposed a graphical-VAR approach that estimates directed paths across time, as well as contemporaneous networks for individuals and between subjects. The VAR approach is an

incredibly powerful model for discovering predictive relations—and hence, potential causal processes—at the level of individuals, but it requires intensive longitudinal data and it typically requires the assumption of stationarity across the course of the time series (i.e., equal means and covariances across all timepoints)

Network models are appropriate to use when it is theoretically plausible that specific behaviors, symptoms, or emotions – measured by individual variables – affect each other directly. Recent applications of network models to psychological data include personality (Cramer et al., 2012), depression (Cramer, van Borkulo, Giltay, van der Maas, Kendler, Scheffer, & Borsboom, 2016; Fried, Epskamp, Nesse, Tuerlinckx, & Borsboom, 2016), schizophrenia (Isvoranu, van Borkulo, Boyette, Wigman, Vinkers, & Borsboom, 2016), quality of life (Kossakowski, Epskamp, Kieffer, van Borkulo, Rhemtulla, & Borsboom, 2016), subjective well-being (Deserno, Borsboom, Begeer, & Geurts, 2016), autism and obsessive-compulsive disorder (Ruzzano, Borsboom, & Geurts, 2014), post-traumatic stress disorder (McNally, Robinaugh, Wu, Gwyneth, Wang, Deserno, & Borsboom, 2014), complicated grief (Robinaugh, Millner, & McNally, 2016), and substance abuse (Rhemtulla, Fried, Aggen, Tuerlinckx, Kendler, Borsboom, 2016). Each of these applications has yielded new knowledge about the potential mechanistic drivers that give rise to psycho(patho)logical constructs. For example, van Borkulo, Boschloo, Borsboom, Penninx, Waldorp, and Schoevers (2015) found that the symptom network of patients whose depression persisted for two years was more densely connected than that of patients whose depression alleviated, suggesting that the persistence of major depression may be related to the strength of symptom connections for individual patients.

In the present paper we continue to use the example of *commitment to school* and *self-esteem* that features in Figure 2. Because it seems theoretically plausible that (a) the items

measuring each construct do not all reflect a single underlying unidimensional construct, and (b) direct causal relations among the attitudes measured by these variables are likely, a network model may reveal new insights into these data. We present a model that can be used to estimate cross-lagged predictive relations among the individual components of these two constructs. In addition to highlighting such predictive relations, the analysis will yield information about which variables in the network are most central to the construct, and about pairs or groups of variables that are tightly clustered and may thus play a unique role with respect to the construct.

Cross-lagged Panel Networks

In the previous sections, we described CLPMs, which highlight pathways among constructs over a few fixed measurement occasions, and network models, which reveal unique pathways among the constituent variables of a construct. We combine these two approaches to form cross-lagged panel *networks* (CLPNs). The key feature of a CLPN is that the relations among individual items are modeled as directed paths across time, reflecting the variance shared between a variable at occasion t and another (or the same) variable at occasion $t + 1$, controlling for all other variables at occasion t . Unlike the undirected networks that are typically estimated on cross-sectional data, paths in the CLPN are *directed* paths from a measurement occasion t to a later measurement occasion $t + 1$. The interpretation of these directed paths is subject to the same constraints as those of the traditional CLPM described above; that is, paths may be interpreted as predictive relations and/or as causal-hypothesis-generating. Similarly, the CLPN cannot reliably estimate differences in predictive patterns across individuals (i.e., every person is assumed to have the same pattern of predictive relations over time) but it can allow different predictive patterns across different measurement occasions (e.g., predictive patterns from 8th grade to 9th grade may differ from those from 9th to 10th grade).

There are strong theoretical and pragmatic benefits to the proposed model. In theoretical terms, the CLPN model can suggest mechanisms by which constructs sustain themselves (e.g., symptoms of major depression that keep influencing one another over time and, as such, result in the persistence of a disorder) and may lead to changes in other constructs over time (e.g., depressive symptoms affecting symptoms of anxiety over time thereby creating the conditions for the emergence of two comorbid disorders). Because a CLPM with latent variables makes the strong assumption that all predictive effects in the observed data are explained by relations among the latent constructs, the latent variable CLPM has no potential to discover which individual components of such constructs have the most strongest influence on longitudinal change and stability. In contrast, the CLPN can uncover the individual behaviors, beliefs, attitudes, traits, or symptoms that are responsible for these auto-regressive and cross-lagged longitudinal construct relations. For example, consider the items *success in school* and *enjoyment of school*, which are both items on the *commitment to school* scale. The CLPN allows that not doing well in school may affect a student's later enjoyment of school over and above that student's mean-level change in school commitment. Moreover, the CLPN allows individual items belonging to one construct to affect change in individual items belonging to a different construct at a later time. For example, not doing well in school may affect a student's later feelings of being a failure over and above the broader effect of school commitment on self-esteem. If individual components of one construct have unique effects on other constructs, the CLPN will highlight these effects. Learning *which* items are primarily responsible for the relations among constructs over time may lead to relevant predictions for how to effect change via intervention.

A major challenge for the CLPN is that there will be many more potential causal paths to estimate and interpret, compared to a traditional CLPM. For example, whereas a CLPM with two latent constructs, each measured by 10 items at 3 measurement occasions, would produce 4 auto-regressive coefficients and 4 cross-lagged paths; a CLPN based on the same data would produce 40 auto-regressive paths (i.e., those connecting each variable at each measurement occasion to the same variable at the next occasion) and 740 cross-lagged paths (connecting each variable at each occasion to each other variable at the subsequent occasion). While it may be possible to estimate 800 regression coefficients, it is not feasible to interpret them individually. The CLPN deals with this problem by (a) using a regularized estimation procedure that fixes the smallest coefficients to exactly zero, thus reducing the number of freely-estimated paths, and (b) computing summary measures such as in-prediction and out-prediction, which reflect the extent to which particular variables (nodes) can be predicted and the extent to which they predict other variables over time, respectively.

The remainder of this paper is structured as follows: In the next section we briefly describe the technical details of our proposed model. We proceed with an empirical example based on data on the constructs *commitment to school* and *self-esteem*. We describe each step of the analysis and offer suggestions for how the results may be interpreted. We consider variations on the model that should be considered, depending on the characteristics of the data and the research question. Finally, we discuss the strengths and limitations of the model more generally, and offer suggestions for its use.

Methodological Details

The process of producing a CLPN model can be broken down into three steps. The first step involves collapsing overlapping items to reduce the set of variables submitted to the CLPN

analysis. The second step involves fitting a series of regularized regression models to estimate the cross-lagged and auto-regressive coefficients across time. Finally, the third step is to summarize the results by producing plots and computing summary statistics such as nodewise in-prediction and nodewise out-prediction. We describe these steps in the following sections.

Collapsing Nodes

The guiding intuition behind network modeling of psychological constructs is that causal power may reside at a much lower level of specificity than the broad constructs that are typically granted causal power (Cramer et al., 2012). But there is no guarantee that the particular items used to assess a construct are at the appropriate level of generality to capture these causal effects. In the case of psychological disorders, variables are typically symptoms, which have a natural ontological status making it reasonable to think about them having causal power. In the case of many cognitive and affective constructs, however, individual variables are typically test or survey items that are chosen to represent diverse aspects of the construct and sets of items may be highly overlapping. While a typical assessment of psychopathology may contain a single item to represent each symptom, a self-esteem questionnaire might include all of the items, “I feel good about myself”, “I am satisfied with myself”, and “I like myself the way I am”. Given a high degree of conceptual overlap, a reasonable expectation is that these three items behave in the same way (i.e., have the same effects and predictors) in the overall network architecture. Thus, we recommend collapsing highly similar items (by summing or averaging scores on these items) to achieve a set of variables that are plausibly autonomous causal agents, thus simplifying the model, reducing the possibility of spurious relations, and rendering the results more interpretable.

It is important to stress that the goal of this analysis step is not to reduce a large set of items to a small set of underlying dimensions, but simply to remove the redundancy that arises from nearly-identical items that query the same manifest behavior. Dimension-reduction techniques such as principal components analysis, singular value decomposition, and factor analysis aim to reduce a large set of items to just a few broad dimensions. This aim is antithetical to the goals of the network approach, which are to understand the connections among low-level behaviors, beliefs, attitudes, abilities, symptoms, or traits. Thus, while it would be possible to use statistical dimension-reduction techniques to decide which items to collapse, we strongly recommend against it. As a central goal of network analysis is to move away from abstractions and toward explaining constructs in terms of concrete observable variables, applying statistical grouping methods is likely to be counterproductive.

We further recommend that researchers err on the side of not collapsing too much, for the same reason: the more variables are collapsed, the more removed the composite gets from a concrete observable behavior/belief/attitude. As the goal is to foster an understanding of the relations among low-level variables, it is important that only variables that are highly similar, that is, variables that are essentially rephrased versions of each other, be collapsed.

Regularized Regressions

Once the data have been condensed into a set of appropriate variables, the next step is to estimate linear regression coefficients of each variable on itself and each other variable, from one measurement occasion to the next. We start with the first two measurement occasions, T1 and T2, according to the model $\mathbf{x}_{T2} = \boldsymbol{\beta}_0 + \mathbf{B}\mathbf{x}_{T1} + \boldsymbol{\varepsilon}$, where $\mathbf{x}_{T1}$ and $\mathbf{x}_{T2}$ are vectors containing all p variables at T1 and T2, respectively, $\boldsymbol{\beta}_0$ is a vector of intercepts for the p variables at T2, $\mathbf{B}$ is a

$p \times p$ matrix of linear regression coefficients² of all p variables at T2 on those at T1, and ε is a vector of residuals for each variable. The model can be extended to more measurement occasions by regressing the variables at each subsequent measurement occasion on the previous occasion.

If the sample size is large enough (at a minimum, sample size must be greater than the total number of variables at all measurement occasions in order for the covariance matrix to be positive definite), the model can be estimated in a single step in an SEM framework³. However, in the following we estimate a series of univariate linear regression models of the form

$x_{j,T2} = \beta_{0,j} + \beta_j' \mathbf{x}_{T1} + \varepsilon_j$, where $x_{j,T2}$ is the j 'th variable at T2, β_0 is its intercept, β_j is the corresponding vector of regression coefficients that predict it from the vector $\mathbf{x}_{T1}$ containing all p variables at T1, and ε_j is its residual. The benefit of the univariate approach is that it allows us to use available regularized regression software to estimate the model, resulting in a more parsimonious model that can be estimated even with small samples and many variables.

Regression coefficients are estimated using lasso regression, which is a type of penalized regression that applies an ℓ_1 penalty on the estimated regression coefficients (Friedman, Hastie, & Tibshirani, 2010). This estimation technique has the effect of shrinking small regression paths to exactly zero, thus achieving a sparse solution. The ℓ_1 penalty is just one possible penalty that can be used to address the problem of overfitting, which is important when estimating a regression with a large number of predictors (the ℓ_2 penalty, i.e., ridge regression, is another possibility). The rationale for using lasso regression is that, on the assumption that some paths are truly zero, it estimates which paths are most likely *not* to exist, in addition to estimating

² We assume that the data are continuous and normally distributed. However, the model can readily be extended to allow for alternative distributions by using generalized linear model with a link function, e.g., a logit link for binary variables.

³ R code to run the simultaneous model is included in the supplementary materials.

coefficients for non-zero paths. The result is a sparse network, in which many of the paths from variables at T1 to variables at T2 will be estimated as exactly zero. Recent simulation results have found that, when sample size is large (e.g., $n = 1,000$), lasso regression typically results in high sensitivity (i.e., edges estimated to exist truly do exist) at the cost of lower specificity (i.e., edges estimated to be zero may or may not actually be zero), but when sample size is small (< 1000) the opposite is true (Epskamp & Fried, 2018; Williams, Rhemtulla, Wysocki, & Rast, 2019; Wysocki & Rhemtulla, 2018). Another feature of regularized regression is that it minimizes overfitting and so its results are more generalizable (i.e., will predict better in a new sample) than ordinary least-squares (OLS) estimates (McNeish, 2015). The regularized

regression estimates are the ones that minimize $\frac{1}{N} \sum_{i=1}^N l(x_{i,j,T2}, \hat{\beta}_{0,j} + \hat{\beta}'_j \mathbf{x}_{i,T1}) + \lambda_j \|\hat{\beta}_j\|_1$, where

$i = 1 \dots N$ denotes individuals. Thus, what is minimized is the sum of individual likelihoods plus a penalty, $\lambda_j \|\hat{\beta}_j\|_1$ in which $\|\hat{\beta}_j\|_1$ denotes the sum of absolute values of the coefficients in $\hat{\beta}_j$, and λ_j is a tuning parameter that determines the strength of the penalty. When λ_j is zero, the penalty drops away and we are left with OLS regression. As λ_j increases, all the coefficients in $\hat{\beta}_j$ will eventually be shrunk to zero. Though there are many criteria that can be applied to choose λ_j , here we use 10-fold cross-validation: Regression estimates are obtained for a sequence of 100 λ values, and the one that produces the lowest cross-validation error is chosen. Recent research has shown that the cross-validation approach to selecting λ_j results in a conservative set of zero-edges; that is, this method results in a less sparse network than other methods (such as EBIC, the most commonly used method in cross-sectional network modeling;

Wysocki & Rhemtulla, 2018). As a result, paths that are estimated to be zero in CV are very likely to be true zeroes, but many paths that are not set to zero may be false positives.

When there are more than two measurement occasions, the regularized regression procedure can be repeated across each successive pair of occasions to generate estimated path coefficients from each occasion to the next.

Cross-time constraints

When more than two measurement occasions are available to model, it may be of interest whether the predictive relations across successive occasions are equal (e.g., if the paths from 7th to 8th grade are equivalent to those from 8th to 9th grade). This question can be answered by fitting two nested models within a SEM framework and comparing their fit. The first SEM model is an unconstrained model that simultaneously estimates auto-regressive and cross-lagged paths across all neighboring pairs of measurement occasions. This model is “unconstrained” because it allows paths to take on different values across subsequent intervals. To make the model more parsimonious, paths that are estimated to be zero in the regularized regression results can be fixed to zero in this step; however, it is important that zeroes be imposed consistently across each pair of measurement occasions in the SEM model to ensure that the next (constrained) model is nested within it. There is no guarantee that the regularized regression will produce a consistent pattern of zeroes across occasions (e.g., the path from variable 1 to variable 2 may be non-zero from T1 to T2, but zero from T2 to T3), so it is necessary to use some decision rule to choose which paths to set to zero consistently across all measurement intervals. A conservative rule would be to constrain a path to zero only if it is estimated to be zero across every neighboring pair of occasions. A more liberal rule would be to constrain a path to zero if it estimated to be zero across the majority of intervals. We recommend using the conservative rule unless there are

good reasons to prefer a more sparse/parsimonious network. Finally, when estimating the full model with multiple measurement occasions, residuals within each measurement occasion should be allowed to covary.

The second (constrained) model is identical to the first but with cross-time constraints imposed, such that the set of paths from T1 to T2 is constrained to be equal to the set of paths from T2 to T3, and so on. Once both unconstrained and constrained models have been fit to the data, they can be compared using a nested chi-square difference test, in addition to approximate fit measures (e.g., CFI, RMSEA) and information criteria (e.g., AIC and BIC, lower values are better). Methodological studies have found that approximate fit indices often out-perform the chi-square difference test when the sample size is large (Cheung & Rensvold, 2002; Little et al., 2007). Based on the results of these fit comparisons, the final selected model may be either unconstrained or constrained.

In-prediction and Out-prediction

The final selected model (either unconstrained across occasions, or constrained, depending on the results of the model comparison in the previous step) can be summarized into two measures of variable centrality: *In-prediction* is the extent to which each variable is predicted by other variables in the network, and *out-prediction* is the extent to which each variable predicts other variables in the network. In-prediction is simply the proportion of variance in each variable at a measurement occasion that is accounted for by the complete set of variables at the previous occasion, and as such it can range in value from 0 to 1:

$$inPred_j = \frac{\text{var}(\hat{x}_{j,T2})}{\text{var}(x_{j,T2})} = \frac{\text{var}(\hat{\beta}'_j \mathbf{x}_{T1})}{\text{var}(x_{j,T2})}.$$

Researchers may be particularly interested in two more specific in-prediction measures: *cross-lagged in-prediction* is the proportion of variance accounted for by all *other* variables at the previous measurement occasion, that is, cross-lagged in-prediction excludes the auto-regressive path, *within-construct in-prediction* is the proportion of variance accounted for by all the variables within the same construct, excluding the auto-regressive effect, and *cross-construct in-prediction* is yet more narrow, it is the proportion of variance accounted for by all variables that belong to a different construct at the previous measurement occasion.

Out-prediction captures the average proportion of variance across all variables at a measurement occasion that is accounted for by a single target variable at the previous occasion:

$$outPred_j = \left(\frac{1}{p} \right) \sum_{k=1}^p \frac{\text{var}(\hat{\beta}_{x_{k,T2}x_{j,T1}} x_{j,T1})}{\text{var}(x_{k,T2})}$$

,

Like in-prediction, out-prediction values can range from 0-1, though they will typically be substantially lower than in-prediction (an out-prediction value of 1 would indicate that a variable accounts for *all* of the variance in *all* variables at the subsequent measurement occasion). As with in-prediction, we can define two narrower version of out-prediction: *cross-lagged out-prediction* averages the predictive effects over all variables except the one that is doing the predicting, *within-construct out-prediction* averages the predictive effects across only those variables that belong to the same construct (excluding the auto-regressive predictive effect) and *cross-construct out-prediction* averages the predictive effects across only those variables that belong to a different construct. Each of these in- and out-prediction measures can be computed from the model estimates produced in the previous step; no new models are estimated.

If the model includes more than two constructs, in-prediction and out-prediction could be computed separately for the effect of each variable on/from variables belonging to each other construct. All measures of in-prediction and out-prediction can be computed for each pair of adjacent measurement occasions. If cross-time constraints have been imposed, then these measures will be identical across occasions. Knowing the stability of an estimate (e.g., its fluctuation in response to sampling variability) is important for calibrating interpretations. Future research should explore how best to accomplish this with in and out-prediction estimates (e.g., bootstrapping the prediction estimates).

Example: Commitment to School and Self-Esteem

To demonstrate estimation and interpretation of a cross-lagged network model, we used publicly available data from the Iowa Youth and Families Project (Conger et al., 2011) on the Commitment to School and Self-Esteem scales. We chose these constructs because they each plausibly feature causal interactions among the attitudes that are measured by individual scale items. For example, the Commitment to School scale measures the extent to which children do well in school (“I don’t do well at school”) and their enjoyment of school (“I like school a lot”), which are likely to affect each other directly.

Data

The Iowa Youth and Families Project collected yearly data from an initial sample of 451 students who were in 7th grade in 1989 until 1992. Commitment to School was measured by seven items developed by Robert Conger for the Iowa Youth and Families Project, and Self-Esteem was measured with the Rosenberg Self-Esteem Scale (Rosenberg, 1965). Participants answered each item on a 5-point scale on which the endpoints corresponded to “Strongly Agree” and “Strongly Disagree”; items were coded such that *lower* scores correspond to higher levels of

commitment to school or self-esteem. Table 1 displays the final set of items that were subjected to the CLPN analysis.

Data on these constructs are available at 4 waves, each separated by one year (1989 – 1992). Of the 451 students who participated in the first wave, 424, 407, and 403 participated in waves 2-4, respectively; 395 students provided data at all 4 waves, 11 of these students were missing data on a single item at one wave. Depending on the analysis, we used listwise deletion (regularized regression analyses with glmnet) or full-information maximum likelihood (SEM analyses with lavaan).

Table 1. Commitment to School and Self-Esteem Items

Commitment to School	
SC1	In general, I like school a lot. (R)
SC2	School bores me.
SC3	I don't do well at school .
SC4	I don't feel like I really belong at school.
SC5	Homework is a waste of time.
SC6	I try hard at school. (R)
SC7	I usually finish my homework (R)
Self-Esteem	
SE1	I feel that I'm a person of worth, at least on an equal level with others. (R)
SE2	I feel that I have a number of good qualities. (R)
SE3	All in all, I am inclined to feel that I'm a failure.
SE4	I am able to do things as well as most other people. (R)
SE5	I feel I do not have much to be proud of.
SE6	I take a positive attitude toward myself. (R)
SE7	On the whole, I am satisfied with myself. (R)
SE8	I certainly feel useless at times.
SE9	I wish I could have more respect for myself.
SE10	At times I think I am no good at all.

Note. Items marked with (R) were reverse-coded.

Analysis and Results

All analyses were done in R (R Core Team, 2016). CLPN regressions were estimated with the glmnet package (Friedman, Hastie, & Tibshirani, 2010) and the lavaan package for

SEM (Rosseel, 2012) , and networks were plotted using the qgraph package (Epskamp et al., 2012). R code to reproduce all analyses and figures is available at osf.io/9h5nj.

Latent Variable CLPM. For the sake of providing a baseline comparison, we begin by presenting the traditional CLPM based on latent variables. We defined two latent variables, Commitment to School and Self-Esteem, each indicated by the set of items presented in Table 1, at each of the four measurement occasions. To ensure that the latent variables are comparable over time, we imposed weak measurement invariance; that is, the unstandardized factor loadings were constrained to be the same at all four measurement occasions. We fit the model in lavaan, using full information maximum likelihood to deal with missing data, and robust corrections to the standard errors and test statistic to deal with violations of the normality assumption (estimator = MLR). The weak invariance constraint did not significantly reduce fit compared to the model that allowed loadings to differ over time, $\chi^2(df = 45) = 47.75, p = .36$. We further constrained the auto-regressive and cross-lagged paths to be constant across each pair of measurement occasions; this model fit significantly worse than the unconstrained model, but an examination of approximate fit indices and information criteria suggested that its fit was not substantially worse, and by some criteria (CFI, BIC) fit was improved by imposing these constraints:

$$\chi^2(df = 8) = 25.00, p = .002, \Delta CFI = -.001, \Delta AIC = 11.46, \Delta BIC = -21.43, \Delta RMSEA = .0001.$$

As is typical in latent variable CLPM analyses, we allowed each item's residual to covary across measurement occasions, to account for stable item-specific variance that is unrelated to the latent factor (Little et al., 2007; Little, 2013; Mayer, 1986). The resulting model with cross-time constraints is depicted in Figure 3 (only two measurement occasions are shown because all path coefficients are identical across time). The model fit was reasonable by conventional

criteria: $\chi^2(df = 2145) = 3984.08, p < .001, RMSEA = .046(.043, .048), CFI = .86, TLI = .85$.

This analysis revealed small cross-lagged coefficients, only one of which was significantly

different from zero at the $\alpha = .05$ level ($SE \rightarrow SC, \hat{b} = .09, p = .046$; $SC \rightarrow SE, \hat{b} = .04, p = .14$

). On the basis of this analysis, one would conclude that the unique predictive effects of school

commitment on Self-Esteem and vice-versa are weak to non-existent.

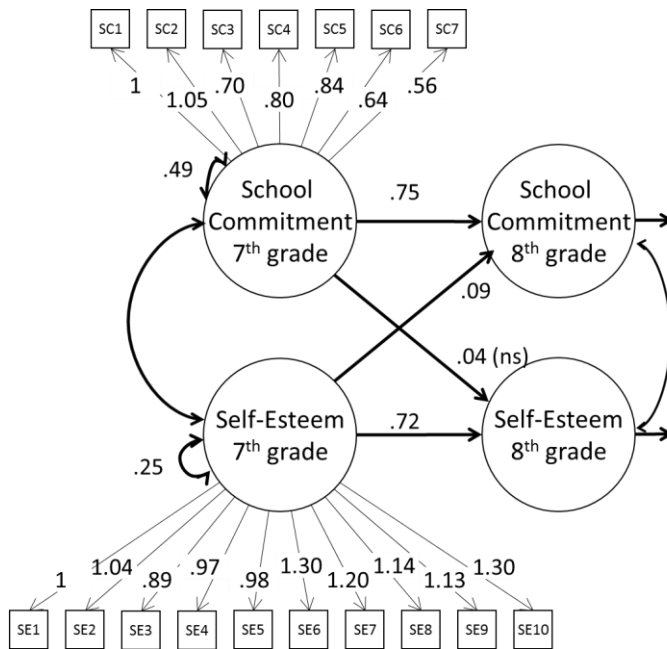


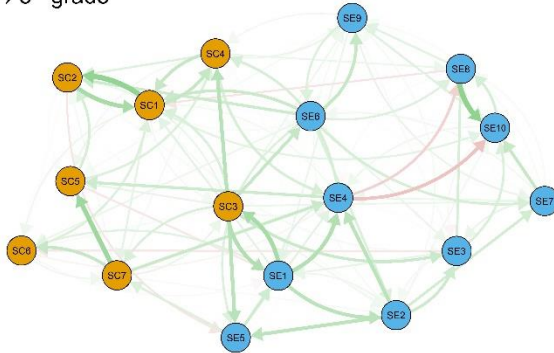
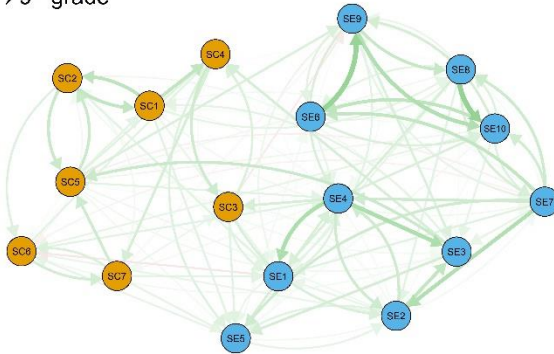
Figure 3. Latent variable CLPM results. For clarity, only the first two measurement occasions are depicted and the measurement model is only shown for the first occasion. All 4 years were included in the model; auto-regressive and cross-lagged paths were constrained to be equal across each pair of occasions. Unstandardized factor loadings were also constrained to be equal at each measurement occasion. Residuals of each item were allowed to covary across occasions (not shown). Residual variances were free to vary across occasions.

CLPN. Next, we followed the steps presented in the Methodological Details section to fit a regularized cross-lagged panel network model. We opted not to collapse any overlapping variables, because (as we argued earlier) this decision should be made on the basis of knowledge about the content domain. Thus, we first fit a series of regularized regressions to derive estimates

of paths from each variable to the entire set of variables for each consecutive pair of measurement occasions (7th to 8th grade, 8th to 9th grade, and 9th to 10th grade). The results of this analysis are plotted in Figure 4 as three directed networks, in which arrows represent cross-time effects (e.g., an arrow from SE4 → SC4 represents a path from SE4 at one measurement occasion to SC4 at the subsequent occasion).

Arrow thickness/darkness represents the relative strength of these effects (thicker arrows represent stronger relations) and color represents the direction of the effect (green arrows represent positive effects). To allow clearer interpretation of the cross-lagged paths, Figure 5 depicts the cross-lagged effects only (a plot that includes the auto-regressive paths, as well as code to produce it, is available in the online supplement at osf.io/9h5nj). The placement of the variables is determined by the Fruchterman-Reingold algorithm described earlier, applied to the first measurement interval (i.e. 7th to 8th grade) and fixed to have the same layout for the subsequent two intervals for ease of comparison.

Similar to the latent-variable CLPM results, the auto-regressive paths (e.g., the effect of SC1 at 7th grade on SC1 at 8th grade, not pictured in Figure 4) tended to be stronger than cross-lagged paths. The auto-regressive paths capture all stability in individual items over time, including construct-related stability as well as any stable unique variance (e.g., a response set that leads participants to answer questions in idiosyncratic ways may be stable across measurement occasions—such an effect would be expected to appear in the auto-regressive paths).

7th → 8th grade8th → 9th grade

- SC1: I like school a lot (R)
- SC2: School bores me
- SC3: I don't do well at school
- SC4: I don't belong at school
- SC5: Homework is a waste of time
- SC6: I try hard at school (R)
- SC7: I finish my homework (R)
- SE1: I'm a person of worth (R)
- SE2: I have a number of good qualities (R)
- SE3: I'm a failure
- SE4: I am able to do things as well as most people (R)
- SE5: I do not have much to be proud of
- SE6: I take a positive attitude toward myself (R)
- SE7: I am satisfied with myself (R)
- SE8: I feel useless at times
- SE9: I wish I could have more respect for myself
- SE10: I think I am not good at all

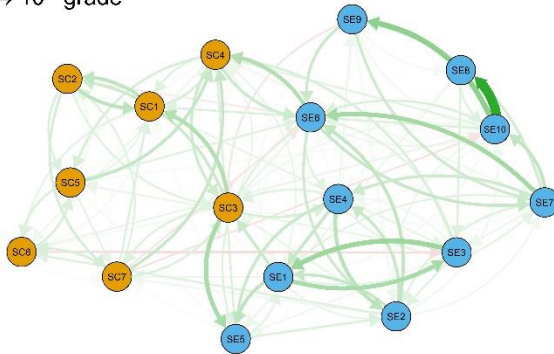
9th → 10th grade

Figure 4. CLPN estimated using a series of ℓ_1 -regularized regression equations. Relations across subsequent measurement occasions were allowed to differ.

The next modeling step is to use an SEM framework to test whether cross-time constraints can be imposed across the 3 measurement intervals. To do this, we first fit an unconstrained CLPN model, where only the paths that were estimated to be zero at all three

measurement intervals were fixed to zero, and every other cross-lagged and autoregressive path was freely estimated. As when we fit the latent-variable CLPM, we again used the full-information estimation method with robust corrections to the fit statistics. This model fit well; $\chi^2(df = 996) = 1453.55, p < .001$; $RMSEA = .032(.029, .036)$; $CFI = .96, TLI = .91$ (robust statistics reported). Second, we fit a model with cross-time constraints imposed, and found similarly good fit,

$\chi^2(df = 1488) = 2085.48, p < .001$; $RMSEA = .031(.028, .034)$; $CFI = .95, TLI = .92$. Because the model with cross-time constraints is nested within the unconstrained model, we can compare the models statistically. The chi-square difference test finds a significant difference,

$\chi^2_{diff}(df = 492) = 635.98, p < .001$. This significant difference suggests that adding the cross-time constraints significantly reduces model fit. However, as the chi-square test is known to be sensitive to large sample sizes, it is useful to also compare fit using approximate measures and information criteria. The constrained model showed a very small reduction in CFI (.014) and a small improvement in RMSEA (.001); moreover, both the AIC and BIC, which weight parsimony in addition to fit, suggested that the constrained model fits better. On the whole, thus, adding the cross-time constraints does not appear to appreciably worsen fit, and the more parsimonious model may improve the generalizability of the results. Thus, we accepted the model with cross-time constraints as our final model.

Figure 5 shows the model with cross-time constraints imposed. Due to these constraints, a single network picture depicts the predictive relations from each measurement occasion to the subsequent occasion. In this figure, paths not significantly different from 0 ($p > .05$) are not shown, nor are auto-regressive paths.

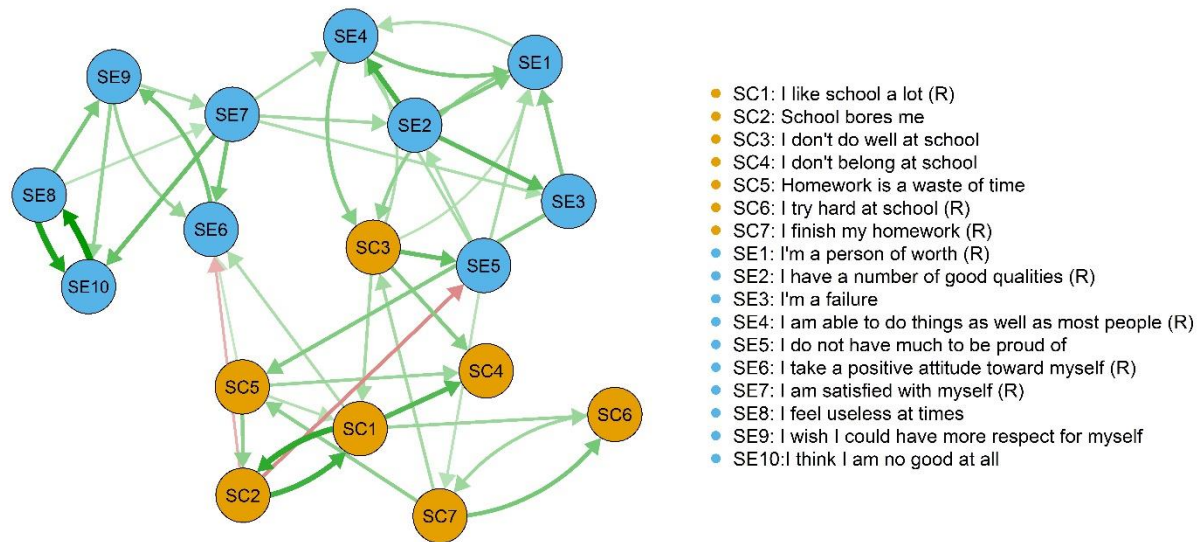


Figure 5. *Estimated CLPN of commitment to school and self-esteem with cross-time constraints imposed. Auto-regressive paths not shown. Paths not significantly different from zero ($p > .05$) not shown.*

Inspection of Figure 5 reveals several interesting cross-lagged paths – impossible to obtain with a traditional CLPM - that may shed light on the ways in which commitment to school can affect self-esteem and vice-versa. For example, school performance (SC3) and self-worth (SE5) appear to be a “bridge nodes” that may connect these two constructs. School performance is predicted by the beliefs that one has many good qualities (SE2) and that one is competent (SE4), and it in turn predicts the belief that one doesn’t have much to be proud of (SE5) and that one is a person of worth (SE1). School performance is also related to other components of commitment to school, for instance by predicting belongingness at school (SC4) and enjoying school (SC1), and being predicted by homework completion (SC7). Self-worth is predicted by school performance (SC3) and interest in school (SC2), and predicts perceived value of homework (SC5) and homework completion (SC5). A model that treats all of these components

as interchangeable markers of a single underlying construct will necessarily miss out on these nuances.

When interpreting the network pathways, it is important to remember that the paths represent conditional predictive relations, that is, the effect of one variable on another, controlling for all other variables at the previous measurement occasion. Remembering this can help to interpret counterintuitive-seeming paths, like the significant *negative* paths from finding school interesting (SC2) to positive self-attitude (SE6) and being proud of oneself (SE5): holding constant other variables in the network (e.g., school performance and enjoyment), students who report being bored with school see themselves more positively than those who do not report being bored. While these negative paths appear to contradict the general positive associations among commitment to school variables (e.g., finding school interesting strongly positively predicts and is predicted by enjoyment of school), they might reflect a tendency for students who see themselves as having mastered the material to feel bored in class.

Finally, we note two pairs of variables that have a strong bi-directional influence on each other: Feeling useless (SE8) and thinking of oneself as no good (SE10), and enjoying school (SC1) and being bored by school (SC2). These pairs of items are good candidates for variables that should likely have been collapsed into a single item before fitting the network models. As mentioned earlier, we elected not to collapse any variables because that is best undertaken with expertise in the content domain. But this result suggests what can happen if that step is skipped, namely that associations between each of these nodes and the remainder of the network may be suppressed. For example, the lack of outgoing paths from SE10 to anything else in the network means that, *controlling for SE8*, SE10 had no further significant predictive relation to any other

variable. If SE8 and SE10 had been collapsed into a single variable, SE8 would not suppress the estimated influence of SE10.

In-prediction and out-prediction indices provide a numerical summary of the influence of individual nodes in the network. Figure 6 displays the cross-lagged in-prediction and out-prediction indices for each variable in the CLPN; these indices reveal the extent to which each variable is predicted by (in-prediction) or predicts (out-prediction) all other variables at the previous / subsequent measurement occasion. Figure 7 and Figure 8 display the within-construct and cross-construct versions of these indices, which limit the set of predictors/outcomes to those from within the same construct (within-construct, Figure 7) or to those from the other construct (cross-construct, Figure 8).

Figure 6 shows that belief in one's competence (SE4) and one's worth (SE1) and perceived positive attitude toward one's self (SE6) are the most highly predictable beliefs (i.e., they have the highest in-prediction), while enjoying school (SC1) and believing oneself to be no good (SE10) are the most useful for predicting other variables in the network (i.e., they have the highest out-prediction). In general, school competence items tend to have lower in-prediction than self-esteem items. Considering within-construct paths only (Figure 7), the patterns of in- and out-prediction look very similar to the overall indices in Figure 6. This makes sense, because most of the edges in the network are within constructs, rather than between. Thus, summary measures of the entire network look similar to the within-construct summaries. In contrast, Figure 8 shows the cross-construct prediction indices, which look fairly different: Here, we see that the commitment to school variables that can be most reliably predicted *by self-esteem variables* are doing well at school (SC3) and having a sense of belonging at school (SC4), and the self-esteem variable that can be most reliably predicted by commitment to school variables is

the feeling of having much to be proud of (SE5). In terms of out-prediction, the most reliable predictors of self-esteem are school performance (SC3), boredom (SC2), and enjoyment of school (SC1), and the most reliable predictors of commitment to school are feeling like a failure (SE3), feeling worthy (SE1), and feeling like one has much to be proud of (SE5). It is important to note here that these measures are simply descriptive summaries of the network, and they do not have standard errors associated with them. As such, we recommend caution in interpreting the relative ranks of these metrics (e.g., a node with the highest out-prediction may not be *significantly* more predictive than any other node).

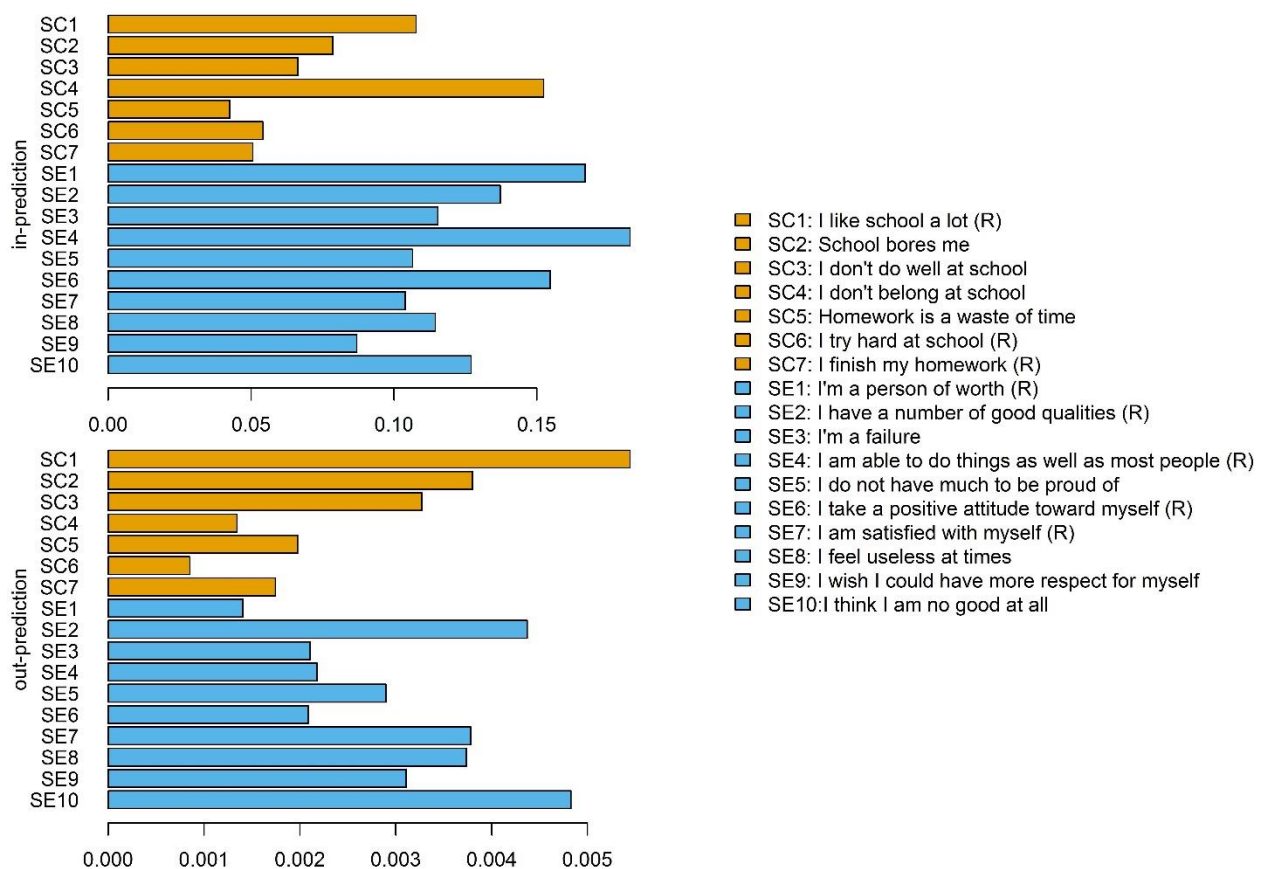


Figure 6. Cross-lagged in-prediction and out-prediction of variables in the estimated CLPN

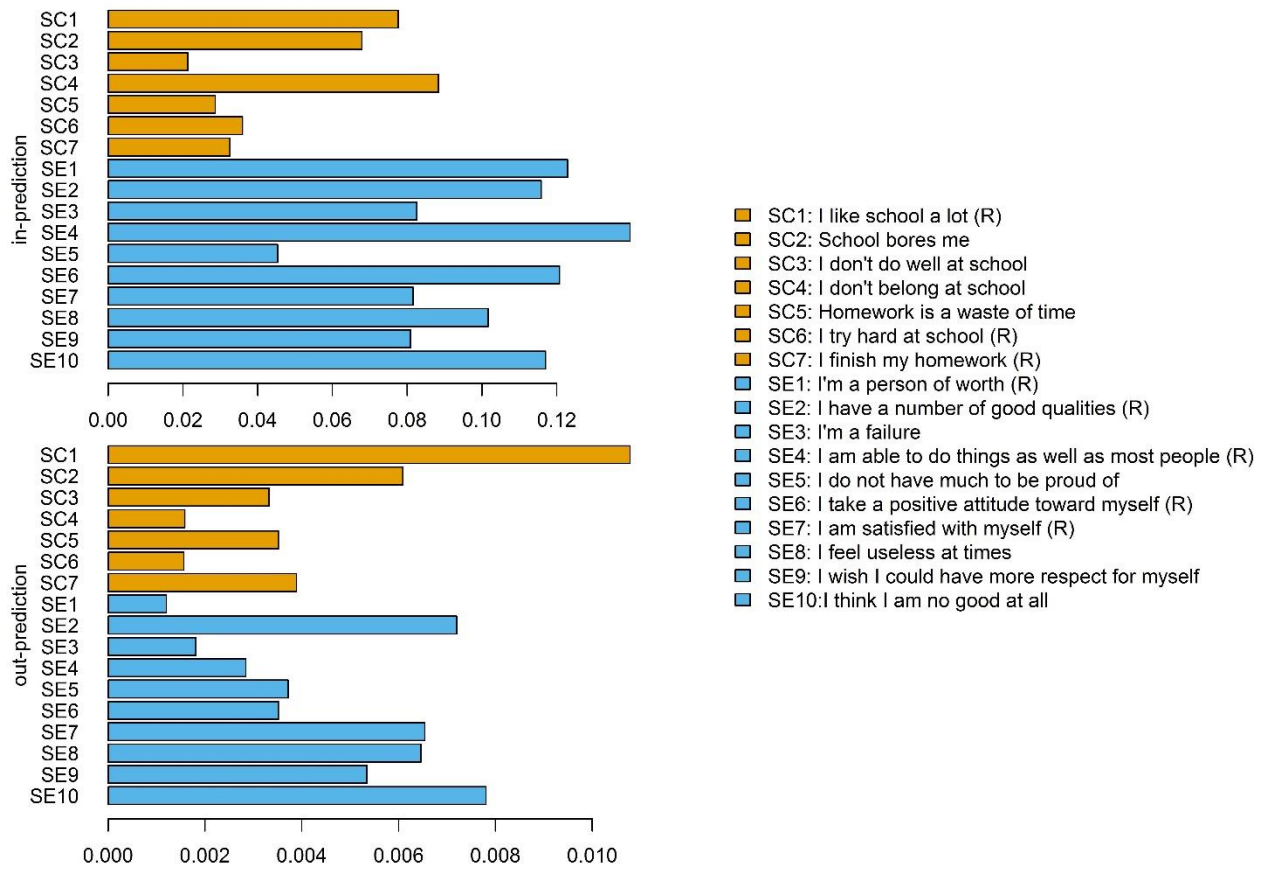


Figure 7. Within-construct cross-lagged in-prediction and out-prediction of variables in the estimated CLPN

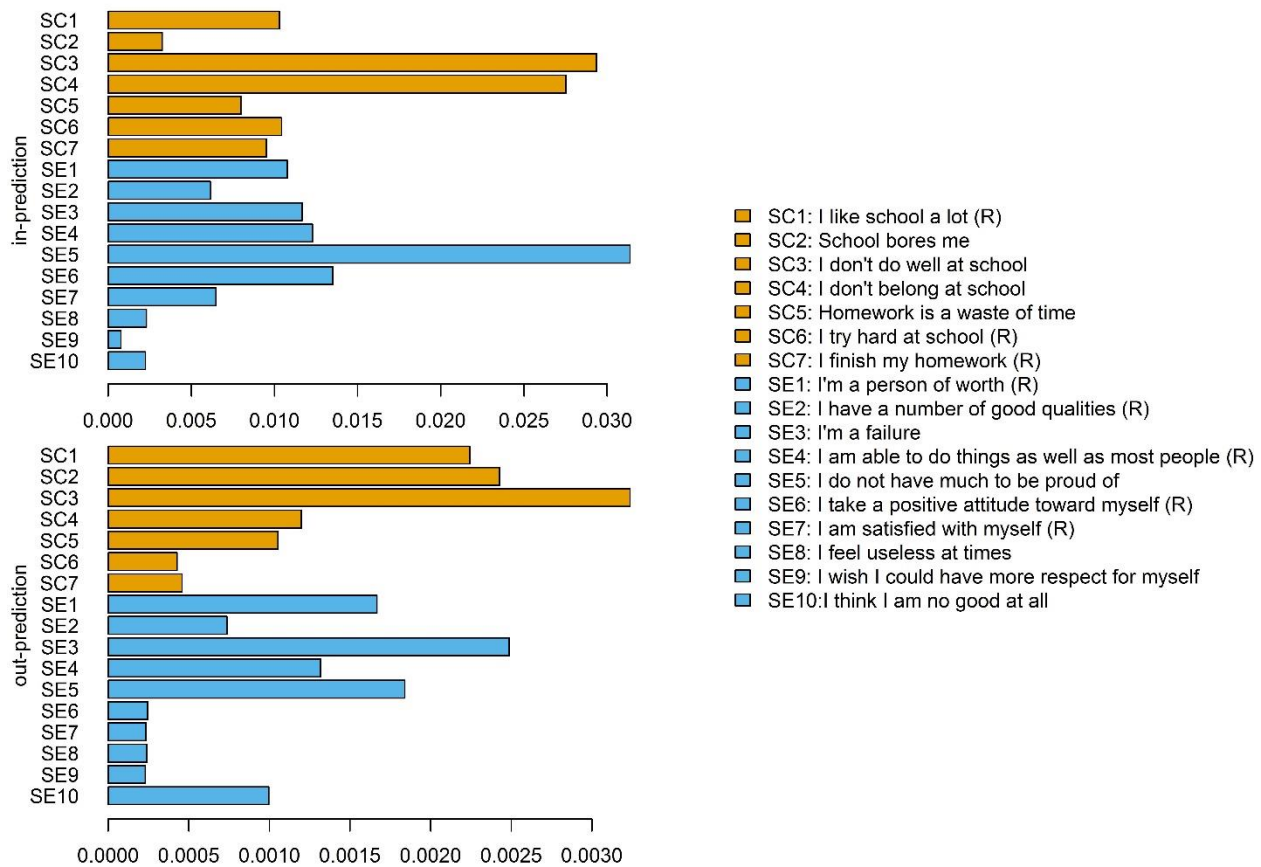


Figure 8. Cross-construct cross-lagged in-prediction and out-prediction of variables in the estimated CLPN.

Discussion

In this paper, we proposed a new model for discovering longitudinal predictive effects of psychological constructs on each other based on data from two or more measurement occasions. The CLPN combines the benefits of latent-variable cross-lagged panel models with the benefits of network representations of psychological constructs to produce a model that can suggest which components of a construct may be responsible for effecting change in another construct, and which components of a construct are most susceptible to change from other constructs. The resulting model may reveal not only the extent to which constructs affect each other, but also much more fine-grained detail that could shed light on the mechanics of these effects.

To be clear, we do not wish to argue that the CLPN is necessarily a better modeling choice than latent-variable CLPM. Both models may provide interesting findings, and one or the other may be more appropriate for a given set of items. In the example we used here, the latent variable CLPM returned non-significant cross-lagged paths among the latent variables, whereas the CLPN produced many potentially interesting item-level effects. In a situation where items are truly reflective of a single underlying construct, we would expect the CLPM to have greater power to show cross-lagged effects, and for the CLPN to reveal few differential predictive patterns among the items. Our goal is simply to add a potentially useful model to the toolbox for cross-lagged panel data.

Challenges and Extensions

Both the traditional cross-lagged panel model and cross-sectional network models are known to have several limitations that also affect CLPNs. We describe some of the major issues here and consider how they might be dealt with.

Time Lag. Estimated autoregressive and cross-lagged regression paths describe effects at a particular time lag, which may not be appropriate for some or all of the effects under investigation (DeBoeck & Preacher, 2016; Cole & Maxwell, 2003; Gollub & Reichardt, 1987). For example, suppose a researcher wants to study the reciprocal effects of school motivation and school achievement. Suppose further that the motivation → achievement pathway takes days or weeks (e.g., motivation is tied to particular assignments, topics, tests), whereas the achievement → motivation pathway happens on the scale of semesters (e.g., school grades from the previous semester affect motivation in the current semester). If waves of data collection are separated by weeks, the researcher might conclude that motivation affects achievement but not vice versa; if instead waves are separated by semesters, the researcher could conclude the opposite. It is thus

essential to choose the time-lag thoughtfully, and to interpret the effects in light of the time-lag represented in the data. A more sophisticated option would be to employ continuous time modeling (Voelkle, Oud, Davidov, & Schmidt, 2012), an approach which, on the assumption that causal effects accrue continuously, produces estimates of causal effects at any arbitrarily chosen time interval. A fruitful future direction for this research would be to extend the CLPN model estimation procedure to allow for estimating the regression parameters in continuous time.

Causation and Mediation. Cross-lagged panel models are frequently used to assess hypotheses about longitudinal mediation effects, but there are two typically untenable assumptions that are necessary to estimate the degree of causal mediation from an indirect path ($X \rightarrow M \rightarrow Y$; Bullock, Green, & Ha, 2010; Fiedler, Schott, & Meiser, 2011; Imai, Keele, & Tingley, 2010; Judd & Kenny, 1981; Pearl, 2012). First, when levels of X are not randomly assigned to participants, it is necessary to evaluate the assumption that, given any observed covariates, X is independent of potential values on M and Y (Imai et al., 2010); which is another way of saying that X is independent of the residuals of M and Y (Bullock et al., 2010). This assumption is untenable whenever any unaccounted-for third variable simultaneously affects levels of X and M , or X and Y .

Second, unbiased estimation of a causal mediation effect requires the assumption that the mediator, M , is independent of potential values of the dependent variable, Y , given the observed values of X and any observed covariates. In other words, M is uncorrelated with any other variable that may influence Y . Unless both X and M are experimentally manipulated and randomly assigned, these assumptions are untestable and typically untenable (Bullock et al., 2010; Judd & Kenny, 1981).

In addition to these assumptions, inferring causality from predictive paths also requires that the measurement time-lag corresponds to the timing of the causal effect (see previous section) and that all the variables are measured without error (see “Measurement Error”, below). As these assumptions, taken together, are typically impossible to evaluate, it is rarely if ever legitimate to infer causality from cross-lagged coefficients.

While experimental manipulation of X and M is the best way to ensure that causal mediation conclusions are warranted, many psychological constructs do not allow for experimental manipulation (e.g., neither commitment to school nor self-esteem can be independently manipulated). In such cases, it is recommended to acknowledge these assumptions and to measure as many omitted variables as possible to control for confounders. In addition, Imai et al. (2010) propose a sensitivity analysis that indicates how large the correlation between error terms would have to be for a mediation effect to disappear.

Ergodicity and Within- vs. Between-Subjects Effects. The CLPN model, like any model based on group-level data, relies on the assumption that participants all come from a single population that is described by a single set of dynamic processes. To the extent that the data contain subgroups that are described by different causal patterns, observed effects will reflect a mixture of processes that may describe no actual individuals (Molenaar, 2004). Put another way, the CLPN model relies on between-subjects covariance to infer predictive processes. For these processes to hold at the level of individuals, all individuals must be assumed to be exchangeable draws from a homogeneous population.

A related issue is that CLPN models conflate between-subject and within-subject variance, which can lead to biased model results when the constructs under investigation contain stable individual differences (Hamaker, Kuiper, & Grasman, 2015; Berry & Willoughby, 2017).

Hamaker et al. (2015) showed that when stable individual differences exist in the constructs being measured, cross-lagged paths conflate these individual differences (between-subjects effects) with the within-subjects effects that researchers are typically interested in. For example, if some people tend to have higher levels of Self-Esteem than others in general (i.e., if people who have higher than average Self-Esteem at T1 also have higher than average Self-Esteem at T2) and if this stable Self-Esteem trait is correlated with a stable Commitment to School trait, then the CLPM or CLPN will produce positive cross-lagged paths among these correlated variables, even if there is no causal effect within individuals.

Several models have been proposed to disentangle these effects (Hamaker et al., 2015; Grimm et al., 2017), all of which require at least 3 measurement occasions. Hamaker et al.'s random-intercept cross-lagged panel model (RI-CLPM) involves fitting a latent factor to the repeated observations, which has the effect of removing the stable individual differences in levels of the trait. The latent factor accounts for between-subjects effects, leaving the within-subjects effects to be revealed in the auto-regressive and cross-lagged paths. Extending the RI-CLPM model to the CLPN would involve fitting a latent factor to each variable in the network simultaneously to modeling the cross-time paths. R code for fitting this model to the empirical data example is included in our online supplementary materials at osf.io/9h5nj.

Measurement error. A common criticism of network models is that they do not account for measurement error. Proponents of network models tend to consider this as a feature, rather than a bug: Unlike latent variable models, which equate all unique item variance (reliable and unreliable) with measurement error, network models allow for the reliable unique variance of each item to be a central component of the causal system. As such, networks use the entire variance of each variable to model a construct, rather than just the part that is shared with all

other items. However, it is undeniably true that random measurement error also exists, and to the extent that observed scores are unreliable, estimated regression coefficients will be biased. One way to improve this situation would be to include multiple indicators (e.g., self- and partner-report) of each behavior, attitude, emotion, or symptom of interest. Each set of items would then be collapsed (i.e., averaged or summed) into a single score before conducting the CLPN analysis. Such an approach would be expected to result in improved reliability of the observed scores while maintaining the network approach's focus on single behaviors, emotions, attitudes, and symptoms as possessing causal power.

Conclusion

Many researchers have access to longitudinal panel data with two or a few fixed measurement occasions on a few key constructs of interest. To address questions of how these constructs affect each other across the time lag of the study, researchers may consider using the CLPN in addition to, or instead of, the traditional CLPM. The CLPN relies on the insight that many psychological constructs do not function as a single causal entity but, instead, their components may have unique causal force. As such, the effects of one construct on another may be due to particular components of each construct. The CLPN can illuminate these specific relations to reveal nuanced and specific effects, and, as such, generate specific, testable causal hypotheses.

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